Sri Lanka Wildlife Conservation Society

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HEC Mitigation

Moragahakanda and Kaluganga Projects of the Mahaweli Authority of Sri Lanka

-INITIAL THOUGHTS-

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Background

In the first Quarter of 2008, SLWCS was requested by both the Department of Wildlife Conservation and the Mahaweli Authority of Sri Lanka to advise them on mitigation of HEC in the MG/KG projects.

SLWCS Initiated surveys utilizing it's own internal funds to develop a comprehensive plan.

The following are preliminary suggestions based on this work, which wasn't as extensive as necersary due to the lack of funding.

Methods

Socio-economic surveys:

- Within Matale and Polonnaruwa districts including all the Divisional Secretariats(DS) and Grama Niladhari Divisions (GNDs) *this is not true. we only covered those within 10km of the boundary of MGKG reservoirs. We did not do anything in Madirigirya which is also necessary. Here I just cut and paste from the online doc. yes only 10 bufferzone*
- systematic surveys were conducted by the SLWCS to assess the intensity and severity of HEC.*how many? two surveys. Socio economic and broad survey*
- Surveys were carried out by interviewing five villagers from each GND including the Grama Niladhari.*how many total GS's and community members? Upul and Chinthaka should answer*
- The socio-economic, demographic and cultural data were collected

- When choosing villages to conduct interviews, priority was given to villages where HEC is significant according to the information gathered from DS offices.

- During the interviews, information such as the farmer's name, cultivated land area, crops, elephant damages to each crops, economic losses, elephant damages to property and injuries to people including death within last few years were collected.

- Secondary data pertaining to HEC

-Grama Niladhari officers - Divisional Secretary officials *?Didn't we get DWC data as well?* *Only through informal discussions..*

- The geographic locations were collected at each survey point using a Global Positioning Systems.

Elephant distribution surveys:

- The surveys were conducted using Broad Reconnaissance Survey method which is widely used in the world to survey large mammals especially elephants.

- This method offers advantages for census of elephants because of the signs of the elephants concentrates along their tracks trails.

- The broad reconnaissance survey method was conducted covering all the habitat types within the project development areas, command areas, resettlement and the potential wildlife habitats.

- When sighting any sign of live elephant, disturbances, land use or habitat type the GPS location were collected using a GPS unit.

- When encounter a sighting such as dung pile, foot print or a feeding sign the age of the sign was also collected along with the GPS location.

The Summary of the Resource Utilization and Socio Economic Survey in Kaluganga

- There are 697 families and 2206 population in the identified GNDs which are affected from the Kaluganga irrigation scheme.

The main ethnic group -Sinhalese

Main religion - Buddhism.

Legal settlements - 94 %

Illegal settlements -6%

- There are 99 households per GND,

- 74 % Structures and 16 % temporary and 9 % mud structures

- 34% of families have access to the electricity and 17% to phone services.

- There is poor local access to healthcare and The Pallegama hospital is the only existed hospital.average distance of 5km from most of villages by road.

- High percentage (71%) of the families has access for the midwife service except Rambukkoluwa and Gangahenwala villagers

- Most of the villages have very irregular and poor transport systems, which in most areas do not existed at all especially in Rambukoluwa Gangahenwala and Madumana.

- 99% of the people have been residing in the same community since their birth and 1% has come as a result of having relationship with other areas.

Economy and land ownership

- The average monthly income - Rs.5965(US$ 55)
- The monthly income of 37% families is ranged between Rs.1000 to 2500.
- The percentage of people involved in farming and daily wage labors is 53%
- Still there are 10% families engage in chena cultivation.
- The highest percentages of families engage in chena cultivation in Karandamulla and Halminiya GNDs (35, 25%)
- The average land ownership is 2.2 acres per family
- Of the up land agriculture produced, 97% is sold, and 83% from paddy

Literacy

- The Literacy level is approximately 89%. But this is badly differs in Rambukoluwa(5%) and Gangahenwala(25%)

- The average of children going school is 104 per G.N.D

- Average distance of 5km for the secondary education.

Organizations

- The Funeral Society (the Maranadhara Samithiya) is the dominant organization. 98% of the households holding memberships.

- The second major institution is the Govi Samithiya, which has 87% membership and provides helps for the farming activities.

- The knuckles range has been declared as the catchment area of the proposed Kaluganga reservoir and 95% of respondents are aware on no entrance or encroachment in these areas.
- 99% of villagers collect and use firewood on a need basis.

HEC

- 82% of the families have faced for the elephant raiding.

Crop damages- 95%

House damages- 3% 2%

Human casualties- 2%

- 2 elephant deaths reported in 2002 and 2006 in Katumanaoya and Rambukoluwa.

- The annual average crop damage area is 6 acres in each village from 2005 to 2007.
- Due to human elephant conflict, Rs.12240 is lost in each village annually.
- According to the villagers Elephant raiding season is mainly from May to October (dry season) in the year.

Perception of the villages on the project

- 76 percent villagers see this irrigation scheme in positively
- The irrigation project is not approved by 16% of people and rest of people has not any idea yet, on the scheme
- 87% families are aware on the project and resettlements and the awareness programs have conducted by Mahawali authority and DS office during 2006-2008
- 80% families know that where the areas to be established for the resettlements
- They request to avoid the isolation of their villages by the project

Summary of the Elephant Research.

- Eastern and North Eastern part of the Moragakanda project area cross over the elephant corridor of the Wasgamuwa National Park to Minneriya- Girithale Nature Reserve and at the Kaluganga project area the corridor spreads along the Victoria- Randenigala toWasgamuwa National Parks
- Population estimates of the elephants based on the fresh dung counts

Kaluganga project area   -50

Moragahakanda project area & RB development area -180

- Around 35 lone bull elephants and the 10-15 herds were ranged in the Moragahakanda project area and the Right-Bank development area.
- Around 12 lone bull elephants and 4-10 herds were ranged in the Kaluganga area according to the fresh dung pile counts.

- Higher number of elephant sightings (Dung counts and feeding signs) were recorded at Attaragallewa, Kirioya, Diggankanda, Diggannawa, Elehera, Punchidigganawa and Thoraapitiya within the Elehera division from July to October 2008.
- Higher number of elephants were recorded in Atharahediya, Kaluganga, Dagiwila, Galgedawala and Kandepitawala areas within Laggala- Pallegama division.

- Dewaladeniya, Thorapitiya, Maoya and Kirioya areas are reportedly have significant HEC circumstances.
- Comparatively in Kambarawa, Galporugalla, Gallaboda, Moragolla, Elagamuwa, Kadawatha, Pubbiliya and Opalgala reported lower abundance of elephants .
- Even though the population of elephants were low in above areas the HEC intensity was high similar to the areas where higher elephant populations discovered.
- This emphasize that the number of elephants in the area not represent the HEC intensity and the HEC intensity highly depends on the problem animals in the areas. The pressure of HEC determine by the problem males and not by the herds.

- Dagiwila, Galgedawala and Atharahediya abandoned villages due to HEC provided enormous variety of food resources for the elephants ranging to the outside from the Wasgamuwa National Park.

- The potential risk of HEC linked with these areas will affect the success of the project because the planned results would not be able to achieve due to HEC.

Habitat preference

- Dung distribution pattern represented the distribution of the elephants, feeding pressure, habitat preferences and the forages preferences. Out of total 567 dung counts that observed 261 dung piles were fresh
- The population in the area showed a special habitat preferences for particular habitat types
- 52.3% of the dung counts were concentrated within the Scrub forest habitat type
- 24.86% of the dung counts were in the grassland habitat type
- 15.34% counts found in the Dry mix or moist monsoon high forests
- 7.5% dung counts concentrated in the Chena or shifting agricultural lands
- But 100% of the observed Chena lands had the signs of the elephants.

- According to above information elephants predominantly range in the scrub type habitats where they access for variety of foods.
- Out of 567 elephant dung counts 313 counts (55.2%) were found vicinity of a water resource.

Issues in relation to Humans and Elephants

1. Loss of habitats, feeding grounds, corridors, migratory routes and home ranges in the proposed reservoirs and the Right- Bank development sites
2. Inappropriate site selection for resettlements where the elephants are ranging outside the protected areas
3. Inappropriate design of corridors for the migration of elephants

1. Proper habitat management plan to enrich the habitats

2. HEC increase in the adjacent areas and the spread towards the Raththota, Naula and Laggala-Pallegama (Eastern) divisions

3. Management of resident elephants in the RB development areas

1. Increase of raiding elephants in the areas
2. Increase the number of orphan elephants in the area
3. Increase the pressure on fauna and flora in the National Park

Summary Suggestion for the HEC Mitigation

- Remnant habitat identification for the elephant conservation must be done as an urgent factor for the habitat enrichment and the habitat management

- Habitat mapping has to be done using the ground truth technology to have a comprehensive understanding regarding the habitat distribution prior to decision making.

- Landscape level habitat management practices has to be done with the advice of the field experts.

- Plant surveys including biodiversity surveys have to be done where the elephant abundance was high to investigate preferred habitats and plant compositions prior to any kind of habitat enrichment project.

- Erect the electric fencing where necessary along with bio fencing
- Research on elephant deterrent /repellent crops
- Promote livestock in the village buffers

Designing of the Corridor

- The corridor that elephant used partially blocked at the Dam construction site right now forever.
- The proposed corridor is the only solution for the elephants that lose their migratory paths.
- Present corridor going through the settlements of Kirioya and Dewaladeniya to Elehera where the HEC in a high intensity.
- The corridor has to broaden from Kirioya to at least Maoya.
- Within the corridor area habitat has to be managed especially for the elephants to avoid the roaming outside from the area.

- Maintain open sparse habitats with plenty of grasslands and scrubs with perennial small waterholes ensure ranging within the protected areas.

- Management of the habitats through control fires and manual clearings as patches and maintain them for a long term will be provided relieve for the displaced elephants.

- Hard boundary needed to be constructed to direct migratory elephants to the Minneriya-Girithale Nature Reserve at Elehera boundary.

- However at least ten elephants from different herds and few lone bull elephants have to be monitored using satellite GPS belts to study the ranging patterns.

Habitat management plan to enrich the habitats

- Plant sampling within the elephant abundant habitats
- Habitat enrichment has to be planned and design according to the plant compositions similar to the elephant abundant habitats.
- New cultivation or introduction of grasses not a viable suggestion for the habitat enrichment.
- Management of existing habitats as cost effective measure
- Expansion of the Minneriya- Girithale Nature Reserve and habitat management
- Complete Biodiversity Studies including Medirigiriya

Renovation of water holes

- Renovation of the tanks should be done within the National Parks not close to the village boundaries, Ex: WNP
- If the tanks are constructed near to the village boundaries elephants tend to gather to the tanks especially during the dry season and will have a trend to enter to the villages and subsequently result to a high intensity of HEC.

Elephant population management

- Elephant drives from the development sites to the protected areas.
- Capture the crop raiding elephants and translocation has to be done to the Flood plains National Park.
- Electric fencing along with biological fencing and effective maintenance around the around the settlements and cultivations,
- Capture the problem animals and translocation to the FPNP and WNP or train them for the cultural activities.

Change income generation

- Introduction of crop diversification instead of traditional monoculture based agriculture.
- Facilitate to develop the livestock as an industry to sustain the economy of the resettles
- Promote and facilitate Agro tourism and Eco tourism

Conservation of elephants

- Ex-situ conservation measures have to be taken with the In-situ conservation measures

- Establishment of a well equipped Elephant orphanage near to WNP
- Promote the ecotourism based on elephants and agriculture

- Conduct the genetic study to estimate the population sizes

- Conduct the GIS based study to get comprehensive idea on elephant ranging patterns

Management of a healthy population of elephants

- Habitat enrichment has to be done to manage a healthy population.

- Otherwise due to high density of elephants and high competition for the feed would result an unhealthy and weak population within few years.

- Habitat management and population management has to be done to have a healthy population of elephants within the protected areas